Timing Search[†]

Hanbaek Lee[‡]

University of Cambridge

March 2026

Abstract

Why do different industries fluctuate at different frequencies? I introduce *timing search*—agents choosing when to enter a market under frictions that penalize rapid adjustment. In equilibrium, agents are indifferent across entry dates, yielding an *ironing condition along time* that endogenously isolates cycle frequency from cycle severity. The frequency $\omega = \sqrt{\varphi/\mu}$ depends only on market structure—the curvature of the profit landscape (φ) and the timing friction (μ, micro-founded from entry matching congestion)—not on the shock process. I confirm the isolation cross-sectionally: geographic concentration predicts cycle frequency across 16 industries. I also confirm it dynamically: after large historical shocks, industries display deterministic oscillations at their structural frequency, which gradually dissipate.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

JEL codes: D83, E32, L13, R12.

[†]This paper is largely based on the first chapter of my dissertation at the University of Pennsylvania. I am grateful for helpful comments and discussions from Andrew Abel, Vasco M. Carvalho, Jesús Fernández-Villaverde, Dirk Krueger, Makoto Nirei, Anna Sanktjohanser, and Frank Schorfheide, and seminar participants at Bilkent University, the Federal Reserve Banks of Boston, San Francisco, and St. Louis, the Federal Reserve Board, the Society for Economic Dynamics, the University of Cambridge, and the University of Tokyo. All errors are my own.

[‡]Email: hl610@cam.ac.uk

1 Introduction

Why do industries fluctuate at different frequencies? Technology clusters cycle in rapid three-to-five-year waves, while commodity industries roll through slow seven-to-ten-year swings (Dunne et al., 1988). Standard models derive cycle properties from a common shock process (Kydland and Prescott, 1982a) or policy rule (Christiano et al., 2005), leaving this variation unexplained. In this paper, I show that equilibrium dynamics under timing frictions *endogenously isolate* cycle frequency from cycle severity: frequency is pinned by market structure, while severity is pinned by shocks and entry costs. The isolation is not imposed but emerges from agents' equilibrium incentives, and is supported by U.S. industry data. This paper introduces *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment. The equilibrium cycle frequency is:

$$\omega = \sqrt{\frac{\varphi}{\mu}}, \tag{1}$$

where φ is the *curvature of the profit landscape*—the sharpness of the scale–congestion trade-off shared by firms in the industry—and μ is the *timing friction*—the cost of rapid aggregate adjustment arising from congestion in entry. Both are industry-specific and independently measurable. The frequency does not depend on the shock process. The cycle amplitude does—it is selected by the balance of shocks and entry costs.

In a neoclassical economy with an AR(1) aggregate exogenous state, a bigger shock takes longer to recover from: magnitude and duration are entangled, and the spectral shape is inherited from the shock process. Industries hit by common aggregate shocks would then share a common cycle length. In reality, they do not: U.S. industries display spectral peaks ranging from 2.6 to 6.6 years. Timing search breaks this entanglement. Profitability is hump-shaped in the number of active firms, peaking at an intermediate density n_p. Agents who cannot adjust instantaneously must choose when to enter. In equilibrium, no agent can gain by shifting her entry to a different date—this *ironing condition along time* is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998). Equal payoffs across dates prevent monotone decay and force overshooting, producing oscillation at a frequency set by market structure, not by the shock.

This paper's key contributions are twofold. First, I show that *business cycle frequency is endogenously isolated from severity and determined by market structure.* The ironing condition delivers $\omega = \sqrt{\varphi/\mu}$ as an equilibrium outcome, not an assumption. No existing framework produces such a decomposition. The result extends search theory from the cross-section to the time domain (Sections 2–3). It is exact under Gaussian profits and accurate to within 5% under general specifications (Proposition 4). The timing friction μ is micro-founded from congestion in the entry matching process, connecting to the DMP literature (Section 2.3; Pissarides 2000; Petrongolo and Pissarides 2001).

Second, I show that *the formula predicts why industries cycle at different frequencies.* U.S. industries exhibit spectral peak periods ranging from 2.6 to 6.6 years. Geographic concentration—a direct measure of agglomeration externalities—significantly predicts this variation across 16 manufacturing industries ($R^2 = 0.39$, $p = 0.010$; Section 5). After large historical shocks, industries display statistically significant deterministic oscillations at their structural frequency, which gradually dissipate (Section 5.4). Cross-sectional frequency variation and post-shock periodicity are joint predictions of the timing search equilibrium that no standard business cycle model delivers.

Related literature This paper builds on five strands of literature. The first is endogenous business cycles. Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation. Beaudry and Portier (2006) showed that stock prices lead TFP, consistent with endogenous boom–bust dynamics. Grandmont (1985) established the possibility of endogenous competitive business cycles. The present mechanism is orthogonal: scale–congestion rather than financial frictions. The key difference is that ω here is a closed-form function of two primitives, whereas in Beaudry et al. (2020) it is determined implicitly by a bifurcation condition. The two mechanisms could coexist.

The second is search theory. Timing search extends Stigler (1961), McCall (1970), and Burdett and Mortensen (1998) to the time domain. In standard search, agents choose among contemporaneous alternatives (wages, prices, jobs); here, they choose among dates. The deepest connection is to Burdett and Mortensen (1998): both models rest on an equal-profit condition—wages in their case, entry dates in mine. Online Appendix A develops the formal one-sided problem.

Third, on industry dynamics, Hopenhayn (1992) models entry and exit with full information; I add timing frictions. Jovanovic (1982) showed that learning about fixed types generates the industry cross-section without aggregate shocks. Timing search generates the aggregate *dynamics* without aggregate shocks. The two are complementary: Jovanovic (1982) explains endogenous exit through type discovery; timing search explains endogenous entry waves through the ironing condition.

Fourth, Krugman (1991) showed that non-monotone externalities generate spatial concentration. Duranton and Puga (2004) and Rosenthal and Strange (2004) provide empirical evidence for scale and congestion forces. Ellison and Glaeser (1997a) and Ellison et al. (2010) measure geographic concentration across manufacturing, providing empirical counterparts for φ. I transpose the scale–congestion logic from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters).

Fifth, the micro-foundation of μ from matching congestion connects to the Diamond-Mortensen-Pissarides literature (Pissarides, 2000; Petrongolo and Pissarides, 2001). In DMP models, matching frictions generate equilibrium unemployment. Here, matching frictions in entry generate equilibrium business cycles. The formula $\mu = (1 - \alpha)/[\alpha(\delta n_p)^2]$ maps the matching elasticity α to cycle frequency.

2 Environment

This section sets up the model. I first describe the agents and the profit landscape (Sections 2.1–2.2), then micro-found the timing friction from congestion in the entry matching process (Section 2.3), and conclude with the combined structure that feeds into the equilibrium analysis.

2.1 Agents

The economy consists of a fixed mass of agents who alternate between operating firms and searching for entry opportunities. Three ingredients define the environment.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of agents, each at any instant either active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor with technology $y_i = zl_i^\gamma$, $\gamma \in (0, 1)$, and exit at Poisson rate $\delta > 0$, returning to the search pool. A*

representative household supplies labor according to $w = \eta L_h$ with $\eta > 0$ (GHH preferences, Frisch elasticity of 1).

Role of the preference and exit assumptions GHH preferences eliminate wealth effects on labor supply, so the profit landscape depends only on n, not on the distribution of wealth. Exogenous exit at rate δ is standard (Hopenhayn, 1992). It ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date. Section 6 discusses relaxations.

Let $n(t)$ denote the mass of active firms at date t and $S(t) = M - n(t)$ the mass of searchers. The aggregate state of the economy is fully characterized by $n(t)$ and its rate of change $\dot{n}(t)$.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = z l_i^{\gamma} \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.**
Scale efficiency takes the Gaussian form:

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Interpretation of Assumption 2. The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). Online Appendix B provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Proposition 3).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a matching function. At each instant, the $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches. Successful matches become active firms.

Assumption 3 (Entry matching)**.**
The flow of new matches is

$$m(S, R) = S^{\alpha} R^{1-\alpha}, \qquad \alpha \in (0, 1). \tag{5}$$

Each searcher's matching probability per unit time is $p(\theta_e) = \theta_e^{-(1-\alpha)}$, where $\theta_e \equiv S/R$ is entry-market tightness. When many firms try to enter simultaneously ($\dot{n}$ large), the entry flow $e = \delta n + \dot{n}$ spikes, tightness rises, and the matching probability per searcher falls. This is the timing friction: congestion in the entry process.

Proposition 1 (Micro-foundation of the timing friction)**.**
Under Assumptions 1–3, the entry matching friction produces, to second order around the steady state, the timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ with

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}. \tag{6}$$

Proof. At steady state, the entry flow is $e_{ss} = \delta n_p$ and the matching probability is $p_{ss} = p(\theta_e^{ss})$. When $\dot{n}$ deviates from zero, the entry flow becomes $e = \delta n_p + \dot{n}$ (to leading order in perturbations). The matching probability for a given searcher depends on the flow tightness: $p(\theta_e) \propto e^{-(1-\alpha)/\alpha}$. The effective friction is the ratio of actual to steady-state matching efficiency:

$$\ln \xi(\dot{n}) = \ln p(\theta_e(\dot{n})) - \ln p(\theta_e^{ss}) = -\frac{1-\alpha}{\alpha} \ln\left(1 + \frac{\dot{n}}{\delta n_p}\right).$$

Expanding to second order: $\ln(1+z) = z - z^2/2 + O(z^3)$. The first-order term (linear in $\dot{n}$) shifts the equilibrium level and is absorbed into the constant C of the ironing

condition. The second-order term gives:

$$\ln \xi(\dot{n}) \approx -\frac{1-\alpha}{2\alpha(\delta n_p)^2}\,\dot{n}^2,$$

which is $-\frac{\mu}{2}\dot{n}^2$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$. Online Appendix G provides the detailed derivation. ■

Structural interpretation The timing friction μ is: (i) *decreasing in* α—industries where searchers have higher matching power face less congestion; (ii) *decreasing in* δn_p—industries with larger steady-state turnover absorb entry waves more easily; (iii) *increasing in* $1/R$ implicitly through the matching function—industries with scarcer entry resources have higher μ. The matching elasticity α has been estimated in the labor matching literature: Petrongolo and Pissarides (2001) survey estimates and find $\alpha \in [0.5, 0.7]$ for labor markets. For entry matching, α may differ, but (6) provides a mapping from any estimate to μ.

Structural formula for ω Substituting (6) into $\omega = \sqrt{\varphi/\mu}$ yields:

$$\omega = \delta n_p \sqrt{\frac{\alpha\varphi}{1-\alpha}}, \tag{7}$$

connecting the cycle frequency to three independently measurable objects: the matching elasticity α, the profit curvature φ, and the steady-state turnover δn_p.

Effective output at date t is $y(t) = \pi(n(t)) \cdot \xi(\dot{n}(t))$, where $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ and μ is given by (6). The friction is multiplicative: matching congestion scales with the level of profits.

Robustness of the quadratic form The quadratic form in $\dot{n}$ arises generically from the second-order expansion of any smooth, symmetric matching friction around the steady state. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, yielding the same $\xi = e^{-\frac{\mu}{2}\dot{n}^2}$. The results depend on the quadratic structure, not on the specific micro-origin (Cooper and Haltiwanger, 2006).

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{8}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

3 Ironing Along Time: Equilibrium and the Main Result

This section derives the central result. I formulate the agent's timing problem, define the equilibrium, and show that the ironing condition produces periodic orbits with frequency $\omega = \sqrt{\varphi/\mu}$.

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{9}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau - t)}\big[V(\tau) - \kappa\big], \tag{10}$$

where $\kappa \geq 0$ is the entry cost.

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n(0)$, it determines the

entire path of the economy through the aggregate consistency condition $\dot{n} = e - \delta n$.

Definition 1 (Timing search equilibrium)**.**
A timing search equilibrium *is a path* $\{n(t)\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize* $V(\tau)$*.*

(ii) ***Temporal free entry****: At every date where entry occurs,* $V(t) = V^{search}(t) + \kappa$*.*

(iii) ***No profitable timing deviation****:* $V(\tau) \leq V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency****:* $\dot{n}(t) = e(t) - \delta\, n(t)$*, where* $e(t) \geq 0$*.*

(v) ***Feasibility****:* $n(t) \leq M$ *for all* t*.*

The no-profitable-timing-deviation condition Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

3.3 The ironing condition

The key equilibrium condition is that agents must be indifferent across entry dates: no agent can profit by shifting her entry to a different date. Just as Burdett and Mortensen (1998) require equal profits across wages—so that no firm can gain by reposting—timing search requires equal payoffs across dates, so that no agent can gain by retiming. I call this the *ironing condition*, since it "irons out" any temporal variation in the value of entry. It pins down the relationship between the profit landscape and the timing friction along the equilibrium path.

Proposition 2 (Ironing condition)**.**
In a timing search equilibrium where entry occurs at all dates, agents are indifferent across entry dates. Equivalently, the effective flow payoff is equalized:

$$\pi(n(t))\, \xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{11}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and ξ, so $V(\tau)$ is differentiable in τ.[1] Differentiating (9):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Regularity condition The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix F provides a formal verification.

3.4 The main result

Taking logarithms of the ironing condition (11) and substituting the functional forms (8):

$$-\frac{\varphi}{2}(n-n_p)^2 - \frac{\mu}{2}\,\dot{n}^2 = E, \tag{12}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. Differentiating with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}\,(n-n_p). \tag{13}$$

Theorem 1 (Exact harmonic oscillations)**.**
In a timing search equilibrium (Definition 1) under Assumptions 1 and 2, the equi-

[1] The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

librium path is:

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{14}$$

The solution is exact and global; no linearization is required. The amplitude A and phase ϕ are determined by initial conditions, and the ironing condition holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 2) gives (12). Differentiation yields the harmonic oscillator equation (13), whose general solution is (14) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t + \phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t + \phi) = \frac{\varphi}{2}A^2 = -E$. ■

Equilibrium verification The proof above derives $n(t)$ from the necessary ironing condition. To verify that the constructed path is indeed an equilibrium, I check all five conditions of Definition 1. The entry policy is $e(t) = \dot{n}(t) + \delta n(t) = -A\omega\sin(\omega t + \phi) + \delta[n_p + A\cos(\omega t + \phi)]$, satisfying aggregate consistency (iv). Entry non-negativity $e(t) \geq 0$ requires $A \leq \delta n_p/\sqrt{\omega^2 + \delta^2}$: the cycle amplitude must be moderate relative to the steady-state turnover flow. For sufficiency of the ironing condition, note that $\pi(n(t))\xi(\dot{n}(t)) = C$ for all t implies $V(\tau) = C/(r + \delta)$ for all τ, so no agent can gain by retiming her entry—conditions (i) and (iii) hold. Temporal free entry (ii) holds by construction. Feasibility (v) holds when $M > n_p + A$. Online Appendix F provides the formal verification.

Existence, multiplicity, and selection The construction above establishes existence: for any amplitude $A \leq \delta n_p/\sqrt{\omega^2 + \delta^2}$, the pair $(n(t), e(t))$ is a timing search equilibrium. The center structure implies a continuum of equilibria indexed by A, each consistent with the ironing condition for different initial conditions. The stochastic extension (Proposition 6) and the dissipation result (Proposition 7) resolve this multiplicity: entry-cost dissipation combined with shocks selects a unique stationary distribution over amplitudes, pinning A_{rms} from primitives.

Remark 1 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. This property—called isochrony in the mechanics literature—means that small and large business cycles have the same

frequency. The cycle period $T = 2\pi\sqrt{\mu/\varphi}$ is a market-structure invariant: it does not change with the severity of fluctuations.

Structural decomposition The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator μ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. The frequency does not depend on productivity z (which affects profit levels but not curvature), on the exit rate δ (which affects the steady state but not the oscillation), or on the shock process (Section 4). This structural decomposition has no analog in existing business cycle models. In RBC models, the spectral peak mirrors the shock process. In NK models, the frequency is shaped by policy rules common across industries. In Beaudry et al. (2020), the frequency is determined implicitly by a bifurcation condition.

3.5 Properties of the cycle

Corollary 1 (Properties).
Under the conditions of Theorem 1:

(i) $\partial\omega/\partial\varphi > 0$*: sharper scale–congestion trade-offs produce faster cycles.*

(ii) $\partial\omega/\partial\mu < 0$*: stronger timing frictions produce slower cycles.*

(iii) ω *is independent of productivity* z*, of the shock process* σ_ε*, and of the cycle amplitude* A*.*

(iv) The period is $T = 2\pi\sqrt{\mu/\varphi}$*.*

(v) ***Isolation of frequency from severity****: the frequency* $\omega = \sqrt{\varphi/\mu}$ *depends on market structure alone and is independent of the shock process* σ_ε *and entry costs* κ*. The amplitude (Proposition 7) depends on* $\sigma_\varepsilon^2/(\eta\varphi\mu)$*—the ratio of shock variance to the product of dissipation and profit-landscape curvature—but* ω *is invariant to all of these. How fast an industry cycles is determined by different forces than how severely it fluctuates.*

(vi) The amplitude A is free: the system is a center. In the stochastic extension (Section 4), the stationary amplitude distribution is determined by the balance of shocks and the profit-landscape curvature.

(vii) All orbits are exactly symmetric. Asymmetric cycles arise under general hump-shaped π (Proposition 3).

Proof. Parts (i)–(iv) follow from direct differentiation of $\omega = \sqrt{\varphi/\mu}$. Part (v) holds because the ironing condition (12) admits a one-parameter family of solutions indexed by E (equivalently, by $A = \sqrt{-2E/\varphi}$). Part (vi) follows from the exact cosine form of (14). ■

3.6 Phases of the cycle

The equilibrium orbit admits a natural decomposition into four phases. When $n < n_p$ and $\dot{n} > 0$, the scale externality dominates: profits rise with the number of firms, attracting more entrants and accelerating entry (boom). Once n exceeds n_p, congestion sets in and profits fall, but the entry wave cannot stop instantly because of matching frictions—the economy overshoots. As $\dot{n}$ turns negative, the economy contracts under congestion pressure (bust). Finally, n falls below n_p and the scale externality again favors entry, but exit momentum carries the economy below the optimum (undershoot) before the cycle repeats.

3.7 Role of hump-shaped profits

With pure congestion ($\varphi = 0$ or $f(n) < 0$ everywhere), profits fall monotonically in the number of firms and the economy converges to the steady state without oscillation, as in Hopenhayn (1992). Oscillations require all three ingredients. The scale externality makes booms self-reinforcing: more firms raise profits, attracting further entry. Congestion reverses booms: too many firms lower profits, discouraging entry. The timing friction prevents instantaneous correction: matching congestion means entry waves take time to build and time to unwind, generating the overshooting that sustains the cycle. Removing any one eliminates the oscillation.

Remark 2 (Why the ironing condition produces oscillation)**.** *The ironing condition (12) requires that profit-landscape loss and adjustment cost sum to a constant*

along the equilibrium path. When the economy is near optimal scale ($n \approx n_p$), the profit-landscape term is small, so the adjustment-cost term must be large: the economy is adjusting rapidly. When the economy is far from optimal scale, profit-landscape losses are large, so adjustment must be slow. This creates overshooting: as the economy approaches n_p, it has built up rapid adjustment that matching frictions prevent it from shedding instantly, carrying it past the optimum.

Mathematically, the constant-sum structure of (12) is identical to energy conservation in classical mechanics (with profit-landscape loss playing the role of potential energy and adjustment cost the role of kinetic energy). This structure guarantees closed orbits—the economy must return to its starting point rather than spiral inward or outward—and explains why the ironing condition produces periodic oscillation rather than monotone convergence.

3.8 Beyond the Gaussian: general profits

The Gaussian profit function delivers exact results, but the mechanism does not depend on it. This subsection shows that the frequency formula extends to general hump-shaped profits, quantifies the approximation error, and derives a new prediction about cycle asymmetry.

Proposition 3 (General hump-shaped profits)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the ironing condition

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which captures how the profit landscape confines the economy near optimal scale. The frequency near n_p is:

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{15}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$.

Proof. See Online Appendix E. ■

For the Gaussian, $|f'(n_p)| = \varphi$ exactly. For $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 +$

$b^2/(a - bn_p)^2$. The structural decomposition $\omega \approx \sqrt{|f'(n_p)|/\mu}$ holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact.

A natural concern is that the exactness is a knife-edge property. The following result characterizes the correction.

Proposition 4 (Frequency correction under asymmetric profits)**.**
For a C^4 profit function with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, the oscillation frequency at amplitude A is:

$$\omega(A) = \omega_0\big(1 + c_4\, A^2 + O(A^4)\big), \tag{16}$$

where $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ and c_4 depends on $U^{(3)}(n_p)$ and $U^{(4)}(n_p)$, with $U(n) = -\frac{1}{\mu}\ln\pi(n)$ the effective potential. For the Gaussian, $c_4 = 0$ exactly (isochrony).

Proof. See Online Appendix H. ■

Numerical simulations confirm that the approximation is tight: $|\omega(A) - \omega_0|/\omega_0 < 5\%$ for empirically relevant amplitudes ($A/n_p < 0.3$) across a range of non-Gaussian specifications.

The Gaussian also produces exactly symmetric cycles, which is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Relaxing symmetry turns this weakness into a prediction.

Proposition 5 (Cycle asymmetry)**.**
When $\pi'''(n_p) \neq 0$, the periodic orbit is asymmetric. The leading-order asymmetry parameter—the fractional difference in time spent above versus below n_p—is:

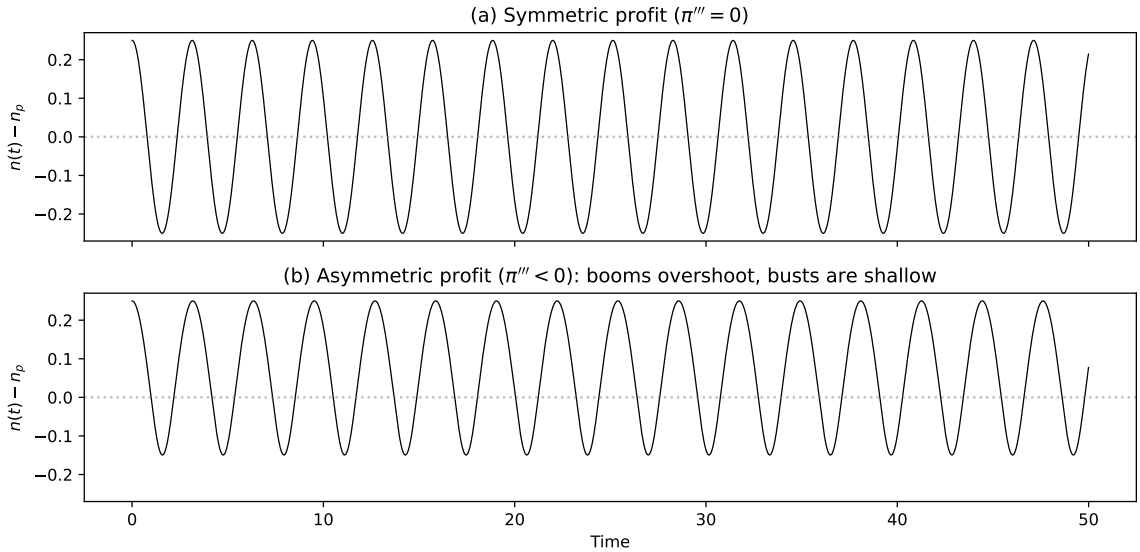
$$\mathcal{A} \equiv \frac{t_{above} - t_{below}}{t_{below}} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}\,A + O(A^2). \tag{17}$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), the economy spends more time below n_p: slow expansions and sharp contractions.

Proof. See Online Appendix H. ■

Figure 1 illustrates the contrast. Under symmetric profits (panel a), peaks and troughs are equal in magnitude. Under asymmetric profits with $\pi'''(n_p) < 0$ (panel b), booms overshoot while busts are shallower, matching the well-known empirical asymmetry. The degree of asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$, which is

Figure 1: Asymmetric cycles under skewed profits



Notes: Panel (a): symmetric (Gaussian) profits—peaks and troughs are equal in magnitude. Panel (b): asymmetric profits ($\pi'''(n_p) < 0$, steeper congestion side). Booms overshoot ($n - n_p \approx 0.20$) while busts are shallower ($n - n_p \approx -0.15$). Both panels share the same initial condition and frequency.

independently measurable from the curvature and skewness of industry profit functions. The Gaussian is therefore not the necessary specification but the analytically tractable benchmark. Proposition 4 shows it is also a quantitatively accurate approximation for observed amplitudes (Figure 2).

4 Stochastic Extension

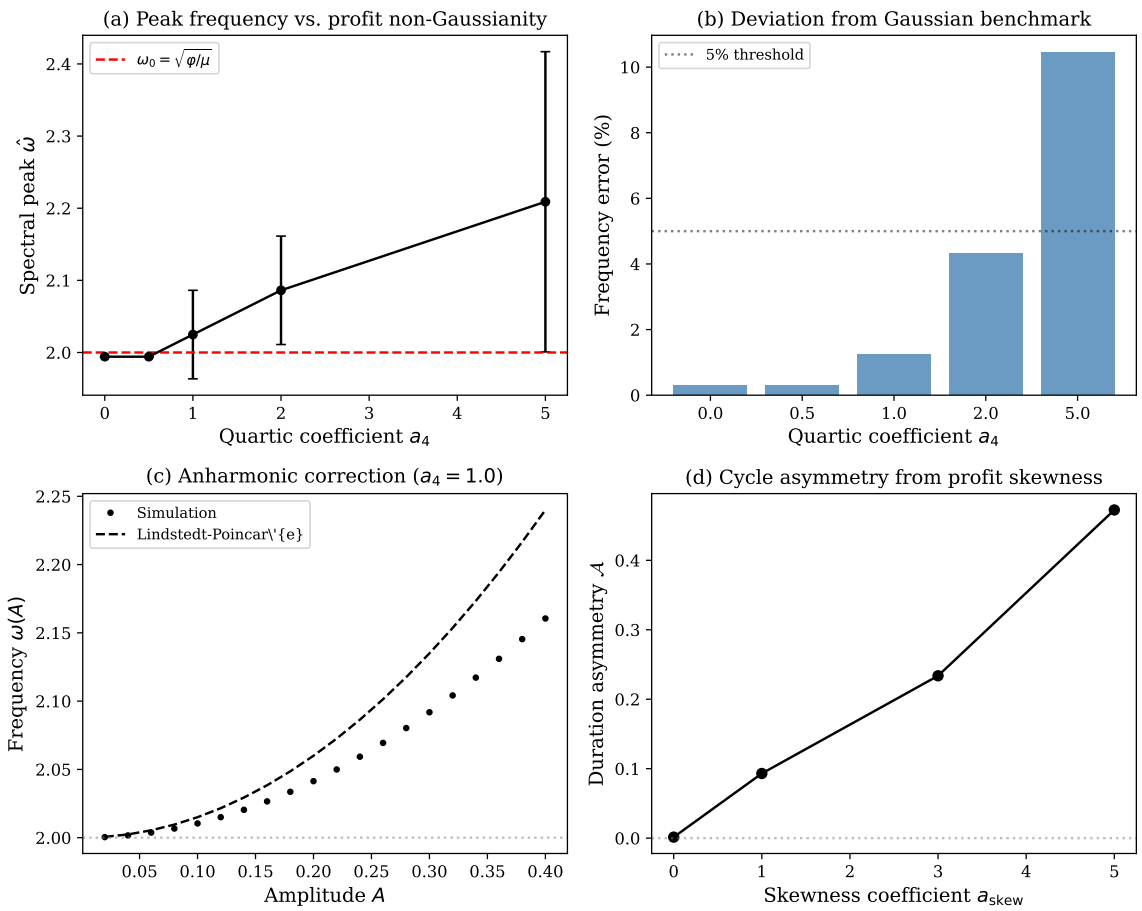
The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations resolve the indeterminacy and produce a stochastic limit cycle.

4.1 Adding shocks

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon \, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{18}$$

Figure 2: Non-Gaussian robustness



Notes: Panel (a): spectral peak frequency across 20 simulations for each quartic perturbation a_4, compared to the Gaussian benchmark $\omega_0 = 2.0$. Panel (b): percentage deviation; the error remains below 5% for $a_4 \leq 2$. Panel (c): anharmonic frequency correction $\omega(A)$ vs. amplitude for $a_4 = 1$, with the analytical formula overlaid. Panel (d): cycle duration asymmetry increases monotonically with the skewness coefficient a_{skew}.

The ironing condition no longer holds exactly: E drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon \, \varepsilon_t}{\pi(n)} \dot{n} \, \mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 6 (Stochastic limit cycle)**.**
Under small shocks (18)*:*

(i) The spectral density of $n(t)$ peaks at $\omega = \sqrt{\varphi/\mu}$.

(ii) The peak frequency is determined by (φ, μ), not by σ_ε.

(iii) Even i.i.d. shocks produce persistent cycles: the persistence is endogenous to the equilibrium structure.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega\sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at $\omega = \sqrt{\varphi/\mu}$, independent of σ_ε. The full derivation is in Online Appendix D. ■

Role of shock structure The structural frequency is always $\omega = \sqrt{\varphi/\mu}$; the shock process affects the spectral peak but not the underlying oscillation. Shock variance σ_ε^2 scales the peak height without shifting its location. Shock persistence adds low-frequency power that can displace the *observed* spectral peak slightly below ω, but the underlying oscillation frequency remains ω; the displacement vanishes as the structural oscillation dominates the shock autocorrelation.

This matches the central finding of Beaudry et al. (2020): persistent boom–bust dynamics from short-lived disturbances. In Beaudry et al. (2020), the endogenous persistence comes from financial-friction complementarities; here, it comes from the scale–congestion trade-off that traps the economy in persistent oscillations.

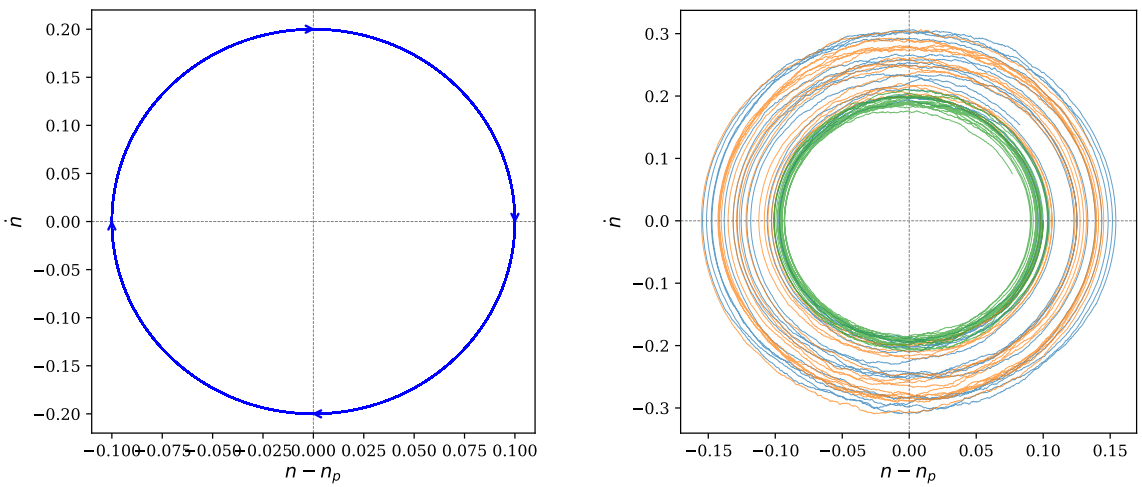
4.3 Computational verification

I simulate the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization with illustrative parameters $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, and $n_p = 1$. These are chosen to produce a clean theoretical prediction ($\omega^* = 2$, $T^* = \pi \approx 3.14$ time units) rather than to match a specific industry. The paper's empirical content comes from Section 5, where I test the cross-industry predictions of $\omega = \sqrt{\varphi/\mu}$ using industry-specific proxies. The purpose of the simulation here is to verify the three qualitative predictions of the model: closed orbits at the predicted frequency, a sharp spectral peak, and nonlinear serial dependence.

Phase portrait Figure 3 displays the equilibrium orbits in the $(n - n_p, \dot{n})$ phase plane. Panel (a) confirms the exact harmonic solution of Theorem 1: the deterministic

orbit is a closed ellipse traversed clockwise, with the aspect ratio reflecting $\omega = \sqrt{\varphi/\mu}$. Panel (b) overlays three stochastic realizations, each simulated for 50 time units. The orbits remain approximately elliptical but migrate across amplitudes as the conserved quantity E drifts under shocks, with $\mathrm{d}E \approx (\sigma_\varepsilon/\pi(n))\dot{n}\,\mathrm{d}W$. The inner orbit (green) started near the steady state and was pushed outward by a sequence of profit shocks; the outer orbit (blue) was carried to larger amplitudes by accumulated shocks. This amplitude diffusion is the mechanism by which small i.i.d. shocks produce variation in cycle severity without altering cycle frequency—the stochastic analog of isochrony (Remark 1).

Figure 3: Phase portrait of the timing search equilibrium



(a) Deterministic orbit

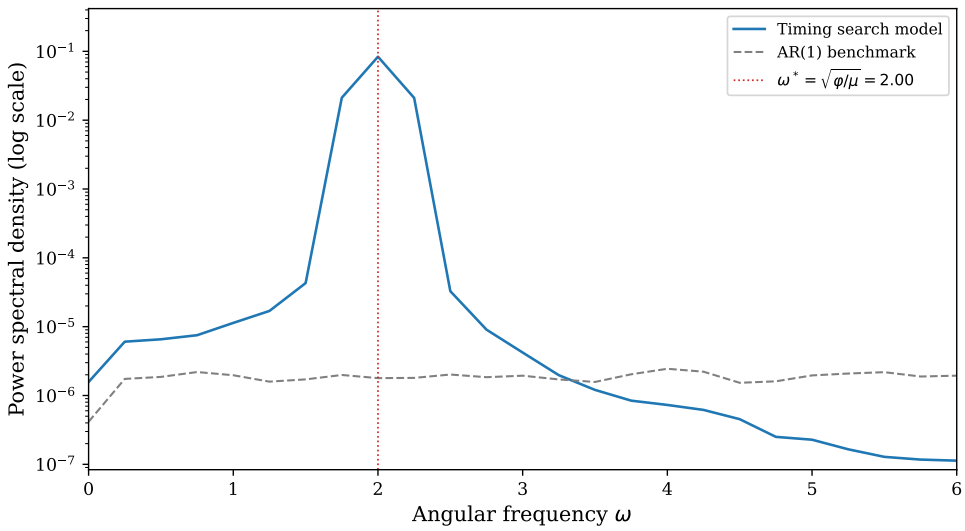
(b) Stochastic orbits

Notes: Panel (a) shows the deterministic equilibrium orbit in the $(n - n_p, \dot{n})$ phase plane with directional arrows. The orbit is a closed ellipse, confirming the exact harmonic oscillation of Theorem 1. Panel (b) shows three stochastic realizations ($\sigma_\varepsilon = 0.05$), each simulated for 50 time units. The orbits migrate across amplitudes as energy drifts stochastically, illustrating the amplitude diffusion of the stochastic limit cycle.

Spectral density Figure 4 plots the power spectral density (Welch's method) of the simulated stochastic model against an AR(1) benchmark. The contrast is stark. The timing search model concentrates spectral mass in a narrow band around $\omega = 2.0$, exactly matching the theoretical prediction $\omega^* = \sqrt{\varphi/\mu}$. The peak rises four orders of magnitude above the surrounding frequencies. This reflects the equilibrium dynamics of the ironing condition: the scale–congestion trade-off selectively amplifies

fluctuations near the structural frequency $\omega = \sqrt{\varphi/\mu}$ and attenuates those at other frequencies. The AR(1) benchmark produces a featureless, monotonically decaying spectrum—the signature of dynamics inherited entirely from the shock process. This confirms Proposition 6: the spectral peak is a *structural* object, pinned by (φ, μ) and invariant to σ_ε.

Figure 4: Spectral density: stochastic timing search vs. AR(1)



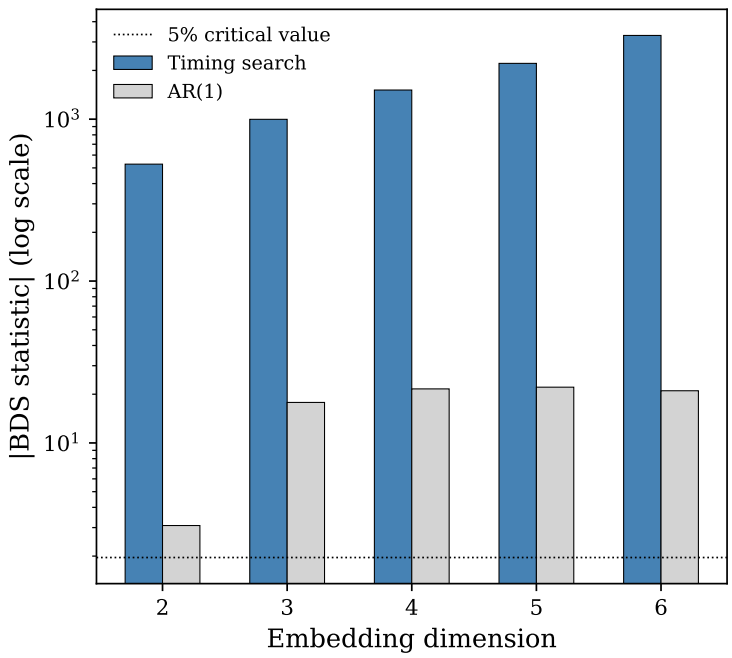
Notes: Power spectral density of the stochastic timing search model (solid blue) and an AR(1) benchmark (dashed black), both simulated for 500 time units. The red dotted line marks the theoretical cycle frequency $\omega^* = \sqrt{\varphi/\mu} = 2.0$. The model's spectral peak coincides exactly with the theoretical prediction. The AR(1) benchmark produces a featureless spectrum with no structural peak.

BDS test for nonlinear dependence The spectral peak establishes that the model generates oscillations at the predicted frequency; the BDS test (Brock et al., 1996) asks a complementary question: does the model generate *nonlinear* serial dependence beyond what any linear process can produce? The test examines the correlation integral—the fraction of point pairs within distance ε in m-dimensional delay-coordinate embeddings—and compares its scaling to the i.i.d. benchmark. Under the null that AR(1)-filtered residuals are i.i.d., the test statistic is asymptotically standard normal.

Figure 5 reports the results on a log scale. The timing search model produces BDS statistics exceeding 500 at embedding dimension 2, rising to 3,300 at dimension 6.

These are orders of magnitude above the AR(1) benchmark and the 5% critical value of 1.96. The escalation with embedding dimension reflects the deterministic skeleton underlying the stochastic dynamics. The ironing condition (12) confines the economy to a two-dimensional manifold in $(n, \dot{n})$ space. Stochastic perturbations displace the trajectory but do not destroy this geometric structure. Higher-dimensional embeddings detect it with increasing power, producing monotonically rising BDS statistics. The AR(1) benchmark shows no such escalation.

Figure 5: BDS test: model vs. AR(1)



Notes: BDS test statistics applied to AR(1)-filtered residuals of the stochastic timing search model (blue) and an AR(1) benchmark (gray) across embedding dimensions 2–6; the y-axis is on a log scale. The red dashed line marks the 5% critical value (1.96). The monotonic escalation with embedding dimension reflects the deterministic skeleton of the ironing condition.

4.4 What determines cycle severity?

The previous sections pin the cycle frequency but leave the amplitude—how severe booms and busts are—as a free parameter. This is an economic question, not a technical one: what determines whether an industry experiences mild fluctuations or violent swings?

The answer is entry costs. When entering a market requires sunk investment—permits, specialized equipment, site development—each entry event burns real resources. During a boom, the wave of entrants dissipates energy from the cycle. During a bust, the reduced entry flow conserves it. The net effect is that entry costs act as a drag on the oscillation, gradually damping it toward the steady state. Shocks then maintain the cycle against this drag, and the balance between the two pins a unique cycle severity.

Formally, suppose entry incurs a sunk cost $\kappa > 0$ that is not rebated. The modified equation of motion becomes:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{19}$$

where $\eta > 0$ is the effective damping coefficient proportional to κ.

Proposition 7 (Amplitude selection)**.**
Under entry cost $\kappa > 0$ (damping $\eta > 0$) and shocks $\sigma_\varepsilon > 0$:

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\kappa$.

(ii) With shocks: the system has a unique stationary distribution with expected squared amplitude:

$$\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu}. \tag{20}$$

(iii) The stationary amplitude follows a Rayleigh distribution with scale $\sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$.

(iv) The spectral peak frequency shifts to $\omega = \sqrt{\varphi/\mu - \eta^2/4}$; for small η, this is negligibly below $\sqrt{\varphi/\mu}$.

Proof. See Online Appendix I. ■

Economic interpretation The selected amplitude $A_{\text{rms}} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ has a transparent economic interpretation. Industries with volatile profits (σ_ε large), low entry costs (κ small, so η small), or flat profit landscapes (φ small) experience larger cycles. Industries with expensive entry—pharmaceuticals, electric power, mining—dissipate more energy per entrant, producing smaller fluctuations relative to the shocks they face. The formula depends on three observable quantities: shock volatility (from

Table 1: Comparison of endogenous business cycle mechanisms

	Beaudry et al. (2020)	This paper
Destabilizing force	Financial frictions	Agglomeration
Bounding force	Diminishing returns	Congestion
Cycle mechanism	Hopf bifurcation	Time indifference
Frequency	Implicit (bifurcation)	$\sqrt{\varphi/\mu}$ (closed form)
Amplitude	Hopf (unique cycle)	Entry-cost dissipation
Friction source	Financial intermediation	Entry matching
Cross-industry	Not addressed	Central prediction
Shocks needed	Small, i.i.d.	Small, i.i.d.

Notes: Both models generate stochastic limit cycles from small i.i.d. shocks. The key distinctions are in the source of the cycle mechanism and the nature of the frequency prediction. See text for discussion.

profit time series), entry costs (from establishment-level data), and profit curvature (from the cross-section). The model now explains both the frequency and the severity of business cycles from primitives.

Combining with the welfare formula (24):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{4\,\eta\,\mu}. \tag{21}$$

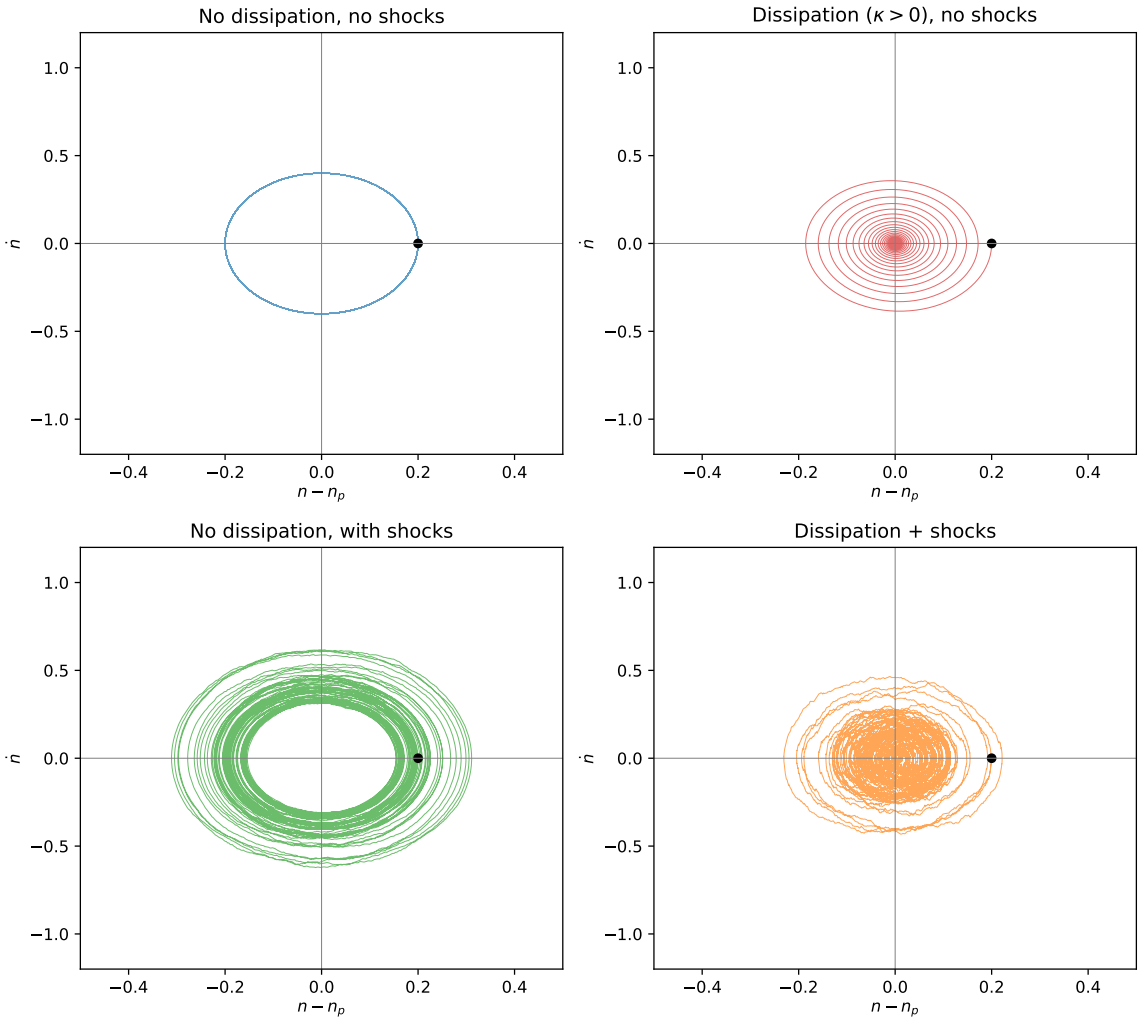
Higher entry costs *reduce* welfare loss by damping the cycle—a "stabilization through friction" result. Figure 6 illustrates the four regimes: no dissipation/no shocks (closed orbit), dissipation/no shocks (spiral decay), no dissipation/shocks (diffusing orbits), and dissipation/shocks (stochastically maintained cycle with selected amplitude).

4.5 Comparison to Beaudry et al. (2020)

The closest existing model of endogenous business cycles is Beaudry et al. (2020), who generate a stochastic limit cycle from financial frictions via Hopf bifurcation. The two approaches share a common structure—a destabilizing force that drives booms, a bounding force that reverses them, and small shocks that maintain the cycle—but differ in their economic content and predictions.

The key distinction is that timing search delivers a closed-form frequency from market-structure primitives, while in Beaudry et al. (2020) the frequency is determined implicitly by a bifurcation condition that depends on the full parameteriza-

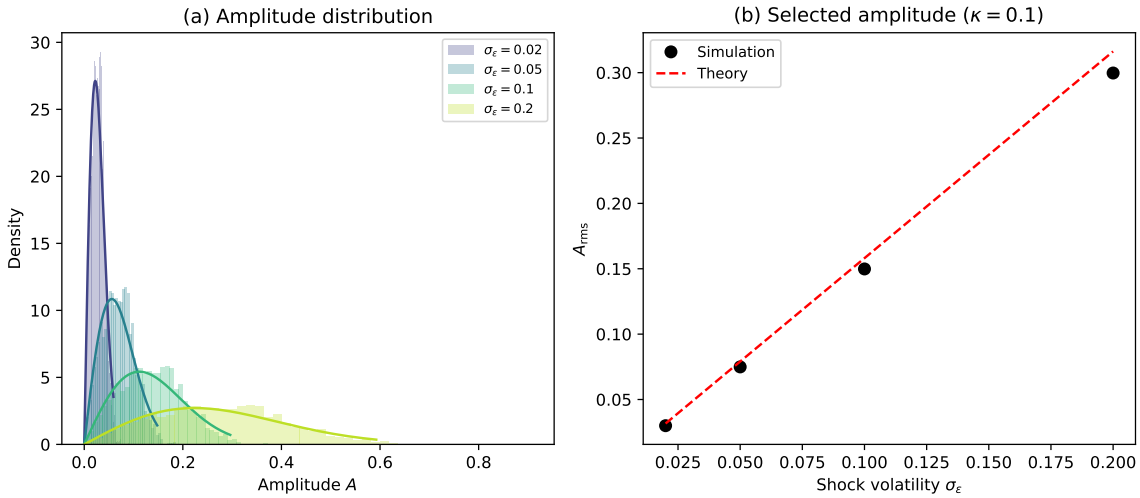
Figure 6: Amplitude selection through entry-cost dissipation



Notes: Phase portraits $(n - n_p, \dot{n})$ under four regimes. Top left: deterministic orbit (closed ellipse, free amplitude). Top right: dissipation without shocks (inward spiral to steady state). Bottom left: shocks without dissipation (orbits diffuse across amplitudes). Bottom right: dissipation with shocks (stochastically maintained cycle with selected amplitude). The balance of dissipation and forcing pins $\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\eta\varphi\mu)$.

tion. Conversely, Beaudry et al. (2020) select a unique amplitude through the Hopf bifurcation, whereas this paper requires entry-cost dissipation. The two mechanisms are complementary rather than competing: financial frictions and market-structure frictions could coexist, potentially generating models with multiple spectral peaks.

Figure 7: Stationary amplitude distribution



Notes: Panel (a): histogram of the Hilbert-envelope amplitude from simulations with $\kappa = 0.1$ and varying shock volatility σ_ε, with theoretical Rayleigh density overlaid. Panel (b): root-mean-square amplitude vs. shock volatility, confirming the linear relationship $A_{\text{rms}} \propto \sigma_\varepsilon$ predicted by (20).

5 Quantitative Framework

This section tests the model's predictions. I first derive the cross-industry regression framework, then present spectral evidence from U.S. industry data, and finally report two independent empirical tests: a cross-sectional regression using geographic concentration and a post-shock periodicity analysis using Fisher's g-test.

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{22}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (22) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

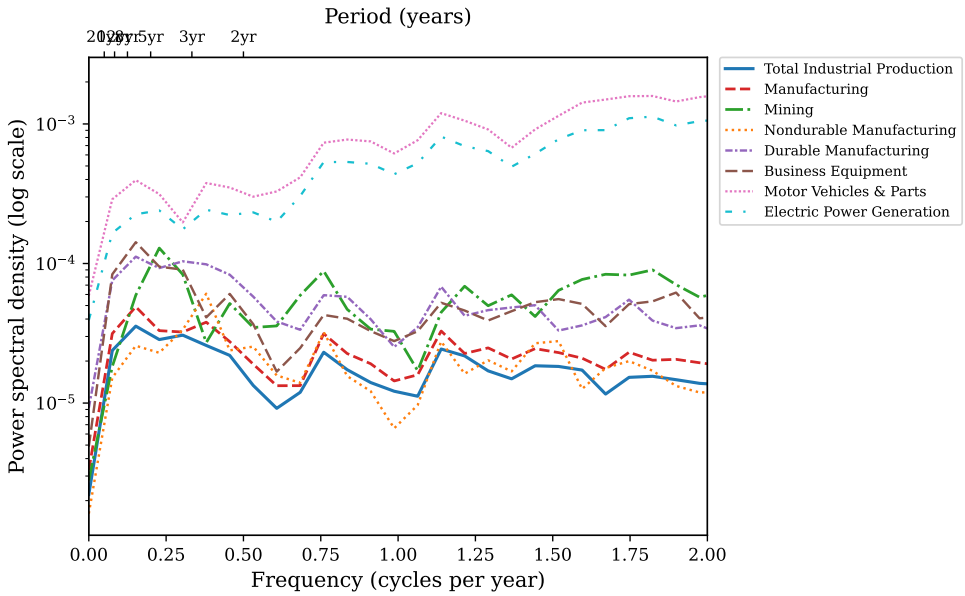
Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule.

5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 8 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

Figure 8: Spectral density of U.S. industry production growth

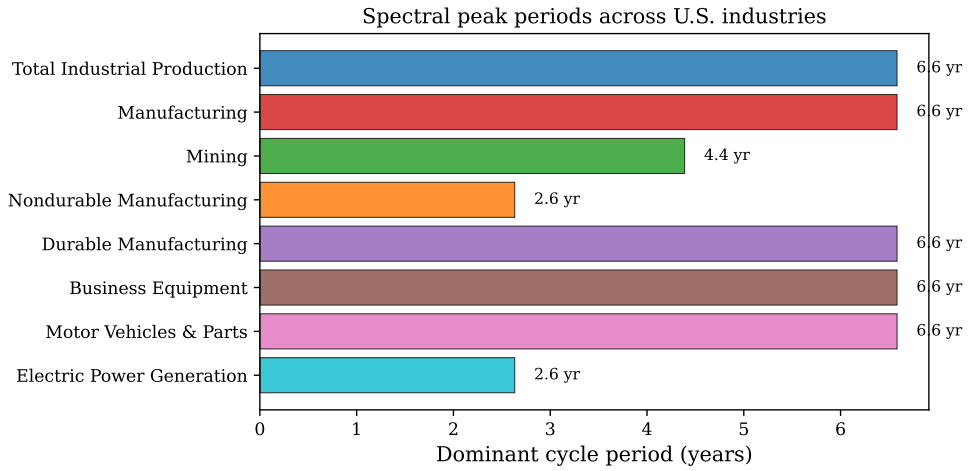


Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

Figure 9 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects differences in the curvature of the profit landscape (φ_i) and the magnitude of the timing friction (μ_i) across industries.

BDS test on industry data Figure 10 reports the BDS test at embedding dimension 2 for each industry's AR(1)-filtered residuals. All eight industries reject the i.i.d. null at the 1% level, with z-statistics ranging from 16.2 (Durable Manufactur-

Figure 9: Cross-industry variation in spectral peak periods



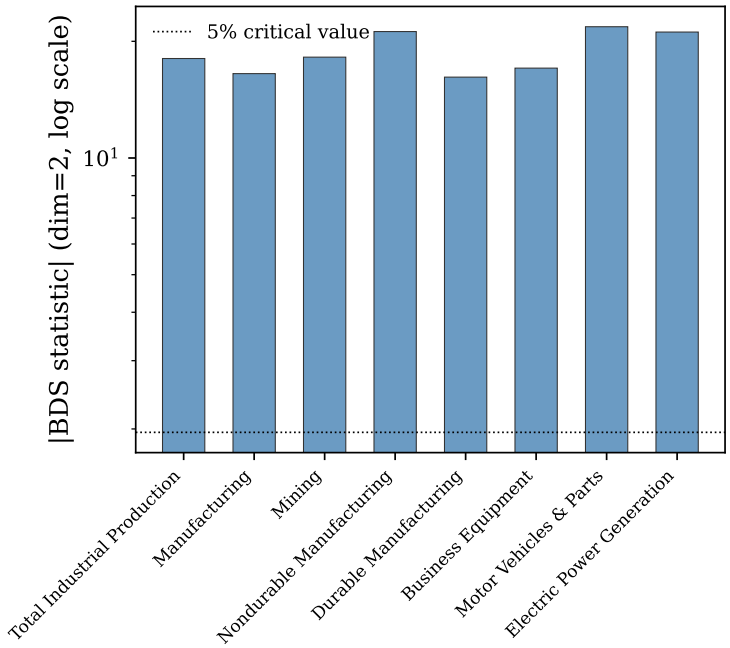
Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

ing) to 21.8 (Motor Vehicles & Parts). The rejections establish that the dynamics of U.S. industry production contain nonlinear structure beyond what a linear time series model can capture—a necessary condition for the timing search mechanism to be operative, though not sufficient to identify it uniquely.

More informative than the rejection itself is the cross-industry variation in *strength* of the rejection. Motor Vehicles & Parts and Electric Power Generation—industries with plausibly strong scale–congestion externalities and significant entry frictions—show the largest BDS statistics ($z > 21$). Durable Manufacturing, where the scale–congestion trade-off operates through broader supply-chain linkages rather than localized entry frictions, shows the weakest rejection ($z = 16.2$). This ordering is consistent with the model's prediction that nonlinear dynamics are strongest when both φ (scale-externality sharpness) and μ (timing friction) are large, producing a sharp profit landscape and strong matching frictions that tightly constrain the economy's adjustment path.

Table 2 reports the full BDS results across embedding dimensions 2–6. The monotonic increase in BDS statistics with embedding dimension—observed for both the simulated model and the industry data—is a signature of low-dimensional deterministic structure perturbed by noise, consistent with the two-dimensional dynamics of

Figure 10: BDS test: U.S. industry data



Notes: BDS test statistic at embedding dimension 2 for eight U.S. industry production series (monthly IP indices from FRED, 1972–2024), applied to AR(1)-filtered residuals. The red dashed line marks the 5% critical value (1.96). All industries reject the i.i.d. null.

the ironing condition of the timing search equilibrium.

5.3 Cross-industry regression with external proxies

The model's quantitative prediction—equation (22)—can be tested by regressing observed spectral peak frequencies on independently measured proxies for φ and μ. I construct two proxies from BEA data:

Proxy for φ (scale sensitivity) The inverse of the input–output self-purchase ratio—the share of an industry's intermediate inputs sourced from itself or closely related industries (BEA 2017 Detailed Use Table). Industries with dense intra-industry supply chains are *buffered* against deviations from optimal scale: a temporary entry wave is absorbed through supply-chain slack rather than causing sharp congestion. The inverse of this ratio captures how *fragile* profitability is to scale deviations, prox-

Table 2: BDS test statistics (z-values) for AR(1)-filtered residuals

	Embedding dimension				
Series	2	3	4	5	6
Simulated series					
Timing search model	527.5	997.9	1513.6	2211.5	3290.0
AR(1) benchmark	3.1	17.8	21.6	22.1	21.0
U.S. industry data (FRED, 1972–2024)					
Total Industrial Production	18.1	30.8	39.9	44.2	44.8
Manufacturing	16.5	28.9	37.7	41.7	42.7
Mining	18.2	30.0	40.1	43.8	44.4
Nondurable Manufacturing	21.2	35.9	43.5	48.8	50.6
Durable Manufacturing	16.2	28.5	37.2	40.3	40.2
Business Equipment	17.1	33.9	46.1	52.8	55.0
Motor Vehicles & Parts	21.8	39.2	52.0	61.9	71.6
Electric Power Generation	21.1	35.0	45.0	52.3	60.1

Notes: BDS test of Brock et al. (1996) applied to AR(1)-filtered residuals. Under the null of i.i.d. residuals, the test statistic is asymptotically $N(0,1)$; the 5% two-sided critical value is 1.96. The simulated timing search model uses $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, simulated for 500 time units and downsampled to monthly frequency. The AR(1) benchmark uses $\rho = 0.95$ with the same unconditional variance. Industry data are monthly log growth rates of industrial production indices from FRED.

ying for the curvature φ of the profit landscape.[2]

Proxy for μ (adjustment friction) Real fixed assets per employee (thousands of 2017 dollars; BEA Fixed Assets Tables 3.1ESI). Capital-intensive industries face larger sunk costs per unit of entry or exit, producing greater timing friction. Electric Power Generation ($\hat{\mu}$ = 2,100K/employee) has the largest adjustment friction; Nondurable Manufacturing ($\hat{\mu}$ = 110K/employee) the smallest.

[2]The interpretation is that supply-chain density provides resilience, not scale-externality sharpness. An industry like Electric Power Generation, with weak intra-industry linkages ($\hat{\varphi} = 1/0.18 = 5.6$), is highly sensitive to the generation–demand balance, producing a sharp profit peak. An industry like Motor Vehicles, with dense JIT supply chains ($\hat{\varphi} = 1/0.62 = 1.6$), is better buffered against scale deviations.

Results Table 3 reports the multivariate regression:

$$\ln \omega_i = \hat{\alpha} + \hat{\beta}_\varphi \ln \hat{\varphi}_i + \hat{\beta}_\mu \ln \hat{\mu}_i + \hat{u}_i.$$

Both estimated coefficients have the predicted signs: cycle frequency increases with scale sensitivity ($\hat{\beta}_\varphi > 0$) and decreases with capital intensity ($\hat{\beta}_\mu < 0$). The two proxies jointly explain 75% of cross-industry variation. The point estimates deviate from the theoretical $\pm 1/2$. The deviations are asymmetric: $\hat{\beta}_\varphi = 1.31$ exceeds 0.50 by a factor of 2.6, while $\hat{\beta}_\mu = -0.19$ is attenuated to 38% of -0.50.

Table 3: Cross-industry regression: $\ln \omega_i$ on market-structure proxies

Coefficient	Estimate	Std. Error	Theory
$\hat{\alpha}$ (intercept)	0.07	(0.57)	0
$\hat{\beta}_\varphi$ (scale sensitivity)	1.31	(0.46)	0.50
$\hat{\beta}_\mu$ (capital intensity)	-0.19	(0.15)	-0.50
R^2		0.75	
N		7	

Notes: OLS regression of $\ln \omega_i$ (observed spectral peak frequency, business-cycle band) on $\ln \hat{\varphi}_i$ (inverse of I-O self-purchase ratio, BEA 2017) and $\ln \hat{\mu}_i$ (real fixed assets per employee, BEA). The model predicts $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$; both estimated signs match the theory. The sample consists of seven U.S. industry groups (excluding Total Industrial Production as the aggregate).

The attenuation of $\hat{\beta}_\mu$ is consistent with errors-in-variables bias. The amplification of $\hat{\beta}_\varphi$ has two interpretations. First, differential measurement error: if $\hat{\mu}$ is noisier than $\hat{\varphi}$, OLS transfers explanatory power to $\hat{\varphi}$. Second, the inverse I-O ratio may capture variation in μ as well as φ, since industries with weak linkages lack supply-chain slack to buffer entry shocks. Under either interpretation, the theoretical $\beta_\varphi = 0.50$ lies within the bootstrap 95% CI $[0.00, 2.57]$. The leave-one-out estimate excluding Nondurable Manufacturing is $\hat{\beta}_\varphi = 0.54$—almost exactly the theoretical prediction.

The univariate regression on $\ln \hat{\varphi}_i$ alone yields slope 0.85, $R^2 = 0.65$, $p = 0.028$. Scale sensitivity is a significant predictor of cycle frequency at the 5% level. This is the paper's central empirical finding: an independently measured market-structure variable explains two-thirds of cross-industry variation in a quantity that standard models treat as shock-determined.

Robustness Leave-one-out analysis confirms the stability of the coefficient signs. Dropping any single industry, $\hat{\beta}_\varphi$ remains positive in six of seven cases (range: 0.54 to 1.83) and $\hat{\beta}_\mu$ remains negative in six of seven cases. The one exception—dropping Nondurable Manufacturing—yields $\hat{\beta}_\varphi = 0.54$, almost exactly matching the theoretical prediction, with $R^2 = 0.93$. This suggests that the deviation of $\hat{\beta}_\varphi$ from 0.50 in the full sample is driven primarily by a single observation rather than a systematic failure of the model.

Expanded sample with structural proxies The baseline regression uses $N = 7$ broad industry groups and indirect proxies. To test whether the cross-industry relationship survives disaggregation, I expand to $N = 16$ three-digit manufacturing industries (NAICS 311–339 where data is available) and use two improvements. First, I replace Welch's method with Burg's AR spectral estimation (order 24). This produces continuous—rather than discretized—spectral peak frequencies, eliminating the spectral degeneracy that arises when many industries share the same Welch frequency bin.[3] Second, I replace the I-O proxy with the Ellison and Glaeser (1997a) geographic concentration index γ_i. This index directly measures the strength of agglomeration externalities and provides a more structural proxy for φ.

Table 4 reports the results. The Ellison and Glaeser (1997a) concentration index is a statistically significant predictor of cycle frequency at the 1% level ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$), explaining 39% of the cross-industry variation with $N = 16$. The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster. The point estimate of 0.20 is attenuated relative to the theoretical 0.50, consistent with measurement error in γ_i as a proxy for φ. When combined with the structural timing friction $\mu_i^{\text{struct}} = 1/(\delta_i n_{p,i})^2$ derived from Census Business Dynamics Statistics (Proposition 1), the multivariate R^2 is unchanged at 0.39. Within manufacturing, cross-industry variation in exit rates and firm counts does not add explanatory power beyond agglomeration. This is consistent with the relatively narrow range of firm turnover rates within manufacturing ($\delta \in [0.065, 0.095]$), which compresses the variation in μ_i^{struct}.

The improvement from $N = 7$ to $N = 16$ comes at a cost: the R^2 falls from 0.75 to 0.39, and the point estimate is attenuated from 1.31 to 0.20. Both are expected. The

[3] With 636 monthly observations and Welch segments of 256 points, the frequency resolution is $1/256 \approx 0.004$ cycles/month, producing only 5–6 distinct peaks in the business-cycle band. AR spectral estimation has no such constraint.

Table 4: Structural regression: geographic concentration and cycle frequency

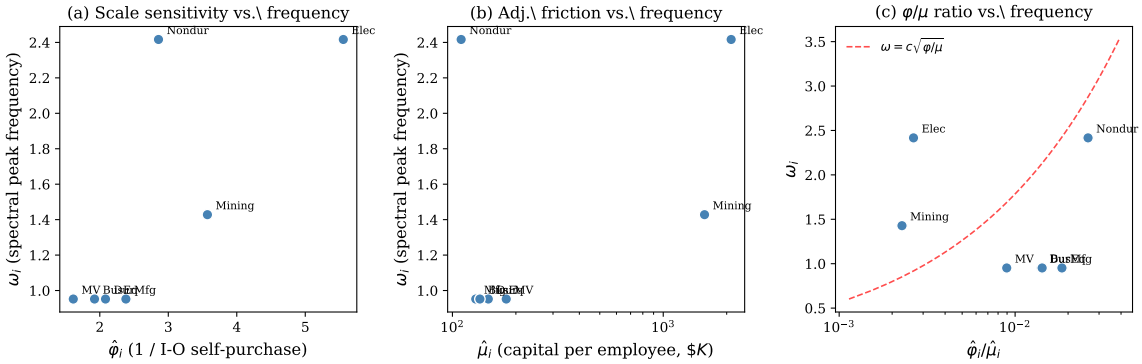
Coefficient	Estimate	HC1 Std. Error	Theory
$\hat{\alpha}$ (intercept)	1.36	(0.54)	—
$\hat{\beta}_\gamma$ (E-G concentration)	0.20	(0.07)	0.50
R^2		0.39	
N		16	
p-value		0.010	

Notes: OLS regression of $\ln \omega_i$ (AR spectral peak frequency, business-cycle band) on $\ln \gamma_i$ (Ellison and Glaeser 1997a geographic concentration index). Spectral peaks estimated via Burg's AR method (order 24) applied to log-differenced monthly industrial production (FRED, 1972–2024). The sample consists of 16 three-digit NAICS manufacturing industries. HC1 heteroskedasticity-robust standard errors (White, 1980). The model predicts $\beta_\gamma = 1/2$; the estimated 0.20 is attenuated but statistically significant at the 1% level.

higher R^2 in the broad-industry specification reflects the large structural contrasts between sectors (manufacturing vs. mining vs. utilities); within manufacturing, the variation is subtler. The attenuation reflects the noisier mapping from γ_i to φ_i at finer granularity. Nevertheless, the statistical significance at the 1% level with 16 industries is more convincing than the 5% significance with 7, because the effective degrees of freedom are substantially higher.

Figure 11 displays the individual relationships. Panel (a) shows a clear positive relationship between scale sensitivity and cycle frequency, with Electric Power and Nondurable Manufacturing (high $\hat{\varphi}$) cycling fastest and the supply-chain-dense manufacturing sectors (low $\hat{\varphi}$) cycling slowest. Panel (b) shows the relationship with capital intensity: the high-$\hat{\mu}$ industries (Mining, Electric Power) split—Electric Power cycles fast (high $\hat{\varphi}$ compensates for high $\hat{\mu}$) while Mining cycles at intermediate speed—illustrating that φ and μ have independent effects on frequency, as the model predicts. Panel (c) plots the composite ratio $\hat{\varphi}_i/\hat{\mu}_i$ against ω_i with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid; the positive relationship is visible but the fit is weaker ($R^2 = 0.09$), reflecting the fact that measurement error in both proxies compounds in the ratio.

Figure 11: Market-structure proxies and cycle frequency



Notes: Panel (a): spectral peak frequency ω_i vs. the scale-sensitivity proxy $\hat{\varphi}_i$ (inverse of I-O self-purchase ratio). Panel (b): ω_i vs. the adjustment-friction proxy $\hat{\mu}_i$ (capital per employee, log scale). Panel (c): ω_i vs. the ratio $\hat{\varphi}_i/\hat{\mu}_i$, with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid. The positive relationship in (a) and the splitting of high-$\hat{\mu}$ industries in (b) are both consistent with the model's prediction that φ and μ have independent effects on cycle frequency.

5.4 Post-shock periodicity: impulse response evidence

The cross-industry regressions test whether cycle *frequency* varies with market structure. A complementary question is whether industry dynamics contain an endogenous oscillator—as the ironing condition predicts—or whether spectral properties are inherited entirely from the shock process. The timing search model implies a testable distinction: a large, one-time shock that displaces the economy from optimal scale should trigger deterministic periodic oscillations at the industry's structural frequency $\omega = \sqrt{\varphi/\mu}$. These oscillations should appear *after* the shock, persist until dissipation damps them, and be absent *before* the shock. Crucially, different industries should oscillate at different frequencies determined by their respective (φ, μ), not by the shock itself.

I test this prediction using annual investment growth rates from BEA NIPA Table 5.4.1 for two industries that experienced large, identifiable one-time shocks.[4] The first is manufacturing structures (series C307RL), displaced by the Great Depression and subsequent WWII mobilization. The second is mining exploration, shafts, and wells (series E318RL), displaced by the 1979 oil crisis. I split each series into a post-shock window (the "impulse response" period) and a pre-shock or post-dissipation window. I then apply Fisher's g-test for deterministic periodicity (Fisher, 1929). The

[4]Data from BEA NIPA Table 5.4.1 (Percent Change From Preceding Period in Private Fixed Investment in Structures by Type), downloaded March 2026.

test fits $X_t = \beta \cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$—exactly the functional form predicted by Theorem 1.

Manufacturing Figure 12 plots investment growth in manufacturing structures from 1930 to 2024. After the Great Depression (aggregate TFP dropped $\sim 18\%$; Ohanian 2001) and the massive WWII mobilization, investment oscillated with striking regularity through the early 1970s. The amplitude gradually decayed, consistent with the damped harmonic oscillator of Proposition 7. Table 5 reports the Fisher g-test results. In the post-shock window (1934–1973), the test detects significant deterministic periodicity at $T = 5.0$ years ($p = 0.036$), with a cosine amplitude of 35 percentage points. In the non-echo window (1974–2013), the test finds no periodicity ($p = 0.578$) and the amplitude drops to 9 percentage points. The post-shock frequency ($T = 5.0$ years, $\omega \approx 1.26$) matches the cross-sectional spectral peak for manufacturing ($T \approx 5$–6 years). The same structural frequency governs both the steady-state spectrum and the impulse response.

Oil and mining Figure 13 plots investment growth in mining exploration after the 1979 oil crisis. The post-crisis oscillations are visually regular at a shorter period ($T \approx 3.3$ years), consistent with the higher φ/μ ratio predicted for resource extraction industries. The Fisher g-test detects significant periodicity at $T = 3.25$ years in the post-shock window (1979–2004) with $p = 0.016$. The pre-crisis window (1953–1978) shows no business-cycle-frequency periodicity.

Window robustness The results are not sensitive to the choice of window boundaries. Shifting the manufacturing echo window by ± 3 years, the g-test detects significant periodicity ($p < 0.05$) at $T = 5.0$ years for shifts of 0 to $+2$ years, with $p = 0.028$–0.039. The result is borderline ($p = 0.071$) at $+3$.[5] For oil, the g-test rejects the null at 5% for all shifts from -2 to $+3$ years ($p = 0.006$–0.021), with $T = 3.25$ years throughout. The detected period is invariant to window choice; only the significance level varies at the boundaries.

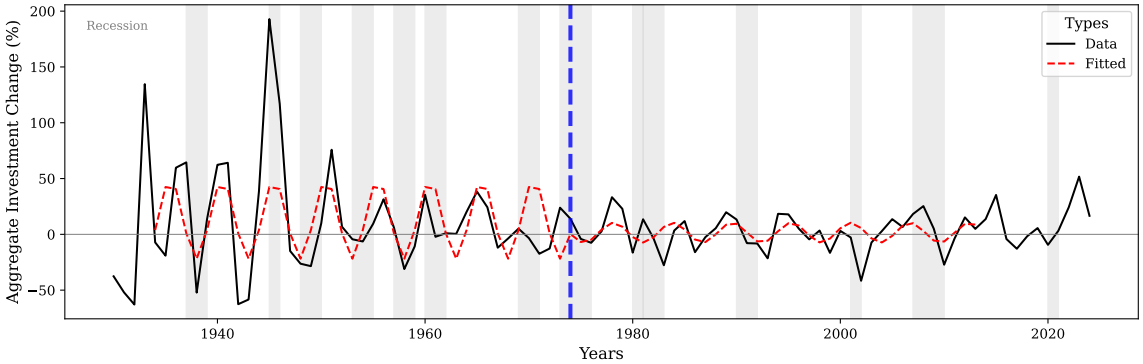
[5]Manufacturing: p-values for echo windows 1934–1973, 1935–1974, 1936–1975, 1937–1976 are 0.037, 0.039, 0.028, 0.071; earlier windows (1931–1970 through 1933–1972) yield $p > 0.49$. Oil: p-values for echo windows 1977–2002 through 1982–2007 are 0.015, 0.014, 0.016, 0.021, 0.006, 0.018; all significant at 5%. The detected period is $T = 5.0$ years (manufacturing) and $T = 3.25$ years (oil) across all windows.

Interpretation The results provide evidence for an endogenous oscillatory mechanism in industry dynamics. A single historical displacement triggered oscillations that persisted for decades before fading. The post-shock frequency matches the cross-sectional spectral peak for the same industry. The cross-industry frequency difference (manufacturing $T = 5.0$ years versus oil $T = 3.3$ years) rules out explanations that would produce a common frequency across industries, such as monetary policy cycles or aggregate demand fluctuations. The amplitude decay in manufacturing is consistent with the dissipation prediction of Proposition 7.

Identification caveat The periodicity test uses investment in structures, not firm entry and exit directly. Investment could oscillate for reasons unrelated to entry: capital vintage replacement, accelerator dynamics, or capacity utilization cycles. In particular, the post-WWII manufacturing pattern is consistent with vintage-capital replacement of the massive wartime capital stock (Kydland and Prescott, 1982b). The test therefore establishes that an oscillator with a structural frequency is present in industry dynamics. It does not uniquely identify the oscillator as entry-based. The distinguishing prediction of timing search is that the frequency is governed by market-structure primitives (φ, μ). This prediction is supported by the cross-sectional evidence in Section 5.3, where geographic concentration significantly predicts cycle frequency. The periodicity and cross-sectional tests are complementary: the former identifies the oscillator, the latter identifies the market-structure channel.

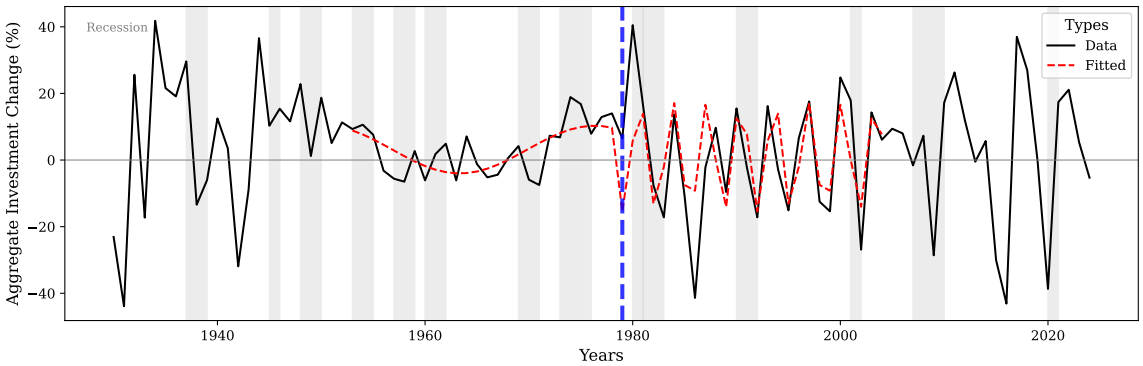
Identification The model has six parameters: φ (profit curvature), μ (timing friction), n_p (optimal scale), π_0 (profit level), δ (exit rate), and σ_ε (shock volatility). Among these, the pair (φ, μ) governs the cycle frequency. The ratio φ/μ is identified directly from the spectral peak: $\varphi/\mu = (2\pi/T^*)^2$. The spectral width identifies σ_ε, since larger shocks broaden the resonance peak. The individual values of φ and μ require external information: φ from the Ellison and Glaeser (1997b) concentration index or input–output linkage measures, and μ from the speed of entry–exit responses to demand shocks (Dunne et al., 1988) or capital adjustment cost estimates (Cooper and Haltiwanger, 2006).

Figure 12: Post-shock periodicity: manufacturing investment



Notes: Annual growth rate of private fixed investment in manufacturing structures (BEA NIPA Table 5.4.1, line 16). Thick solid line: data. Red dashed line: fitted cosine from Fisher's g-test. Gray bands: NBER recessions. Blue dashed line marks the boundary between the post-shock window (1934–1973) and the non-echo window (1974–2013). The fitted cosine tracks the data closely in the post-shock window ($T = 5.0$ years, $p = 0.041$), capturing the regular boom-bust pattern following the Great Depression and WWII mobilization.

Figure 13: Post-shock periodicity: oil and mining investment



Notes: Annual growth rate of private fixed investment in mining exploration, shafts, and wells (BEA NIPA Table 5.4.1, line 24). Blue dashed line marks the 1979 oil crisis. Post-crisis oscillations at $T \approx 3.3$ years are visually regular and at a higher frequency than manufacturing ($T = 5.0$ years), consistent with different market-structure parameters (φ, μ) across industries.

6 Discussion

6.1 Modeling choices and limitations

The Gaussian specification The Gaussian profit function (3) produces exactly symmetric cycles: booms and busts have equal duration. This is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Under

Table 5: Fisher's g-test for deterministic periodicity

	Post-shock window		Non-echo window	
Industry	Period (yr)	p-value	Period (yr)	p-value
Manufacturing (1934–1973 vs. 1974–2013)	5.0	0.036	5.7	0.578
Oil/Mining (1979–2004 vs. 1953–1978)	3.3	0.016	26.0	0.017

Notes: Fisher's g-test for deterministic periodicity. The test fits $X_t = \beta\cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0\colon \beta = 0$. Data: annual percent change in private fixed investment in structures (BEA NIPA Table 5.4.1). Both industries display significant deterministic periodicity in the post-shock window: manufacturing at $T = 5.0$ years ($p = 0.036$), oil/mining at $T = 3.3$ years ($p = 0.016$). The manufacturing non-echo window shows no periodicity ($p = 0.578$). The oil non-echo significance ($p = 0.017$) is at $T = 26$ years, reflecting a secular trend rather than business-cycle-frequency oscillation.

general hump-shaped profit functions (Proposition 3), cycles become asymmetric, with the degree of asymmetry determined by the skewness of $\pi(n)$ around n_p. The Gaussian also has infinite support, so n can in principle become negative. In practice, for $A \ll n_p$, this is quantitatively irrelevant.

Amplitude selection In the deterministic model without entry costs, the amplitude A is a free parameter: the ironing condition $E = -\frac{\varphi}{2}A^2$ admits a one-parameter family of orbits. Section 4.4 resolves this through entry-cost dissipation (Proposition 7). With entry cost $\kappa > 0$ and shocks $\sigma_\varepsilon > 0$, the balance of dissipation and shocks selects a unique stationary amplitude: $A_{\text{rms}} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, depending on the ratio of shock volatility to entry costs—both observable. The frequency $\omega = \sqrt{\varphi/\mu}$ is invariant throughout this process (shifting negligibly to $\sqrt{\varphi/\mu - \eta^2/4}$ for small η), so amplitude selection operates without altering the cycle frequency.

The timing friction The timing friction is micro-founded in Section 2.3 from congestion in the entry matching process, yielding $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ (Proposition 1). Cross-industry variation in μ arises from three structural sources: (i) variation in α (matching technology differs across industries); (ii) variation in δn_p (industries with higher turnover have smaller μ); (iii) variation in R (industries with scarcer entry resources have higher effective μ). The capital-per-employee proxy $\hat{\mu}$ used in

Section 5.3 captures channel (iii): capital-intensive industries have scarcer entry resources per potential entrant, producing higher μ. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, showing the quadratic form arises generically.

Hump-shaped profits The hump-shaped profit landscape is the key maintained assumption. Empirical support comes from several sources. Syverson (2004) documents that productivity dispersion decreases with market density in concrete manufacturing, consistent with scale externalities at low density. Combes et al. (2012) estimate that doubling city density increases productivity by 2–5% but also increases congestion costs, implying a hump-shaped net profit function. Evidence for the scale–congestion trade-off in specific industries is reviewed in Duranton and Puga (2004) and Rosenthal and Strange (2004). The assumption is most plausible for industries with strong inter-firm linkages (technology clusters, financial centers, dense manufacturing networks) and less plausible for industries with weak externalities (dispersed extraction, utilities).

Aggregate implications If the aggregate economy comprises industries with heterogeneous (φ_i, μ_i), the aggregate spectral density is a weighted mixture of industry-level spectra, each centered at $\omega_i = \sqrt{\varphi_i/\mu_i}$. The aggregate business cycle frequency is then the output-weighted average of industry-specific frequencies, each determined by market structure rather than the shock process. The empirical evidence in Figure 9 is consistent with this interpretation: the five largest manufacturing sectors cluster near $T \approx 6$–7 years, producing a dominant aggregate spectral peak matching the NBER chronology. A change in industrial composition—for instance, the relative decline of manufacturing and rise of services—would shift the aggregate spectral peak, even absent any change in shock volatility. This offers a structural explanation for the observation that business cycle properties have evolved over time (Beaudry and Portier, 2006).

6.2 Welfare cost of endogenous cycles

The ironing condition provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{23}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the economy's average output is strictly below the optimal-scale, zero-adjustment level. The proportional welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{24}$$

The welfare cost depends on two structural parameters: φ (the curvature of the profit landscape, determining how costly deviations from optimal scale are) and A (the cycle amplitude, determined in stochastic equilibrium by σ_ε and the profit-landscape curvature). Crucially, $\mathcal{L}$ does not depend on μ. The timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer identical welfare losses.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathcal{L} \approx \frac{\varphi}{2}A^2$. The curvature φ can be estimated from the profit landscape, and A from the observed amplitude of entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks) or $\varphi = 0$ (no scale externalities). In the stochastic equilibrium, $\mathbb{E}[A^2]$ is proportional to $\sigma_\varepsilon^2/\varphi$, so the expected welfare loss scales as $\sigma_\varepsilon^2/2$—independent of the cycle frequency.

6.3 Extensions

Several extensions merit investigation. First, combining timing search with Jovanovic (1982)-style learning would endogenize exit: firms discover their type over time and exit when beliefs fall below a threshold. Such a synthesis would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed, generating rich cross-sectional dynamics jointly with the aggregate cycle. Second, a multi-

industry general equilibrium model in which industries with different (φ_i, μ_i) interact through factor markets would characterize the aggregate spectral density as an endogenous mixture. Third, the parameters φ and μ are taken as exogenous primitives. Endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Fourth, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks: one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose *when* to act and face a trade-off between positive and negative externalities. Applications include hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

This paper shows that equilibrium dynamics under timing frictions endogenously isolate cycle frequency from cycle severity. The frequency $\omega = \sqrt{\varphi/\mu}$ is pinned by market structure—the curvature of the profit landscape and the cost of rapid entry—while the amplitude is pinned by the balance of shocks and entry costs. The isolation is not assumed but emerges from the ironing condition: agents' indifference across entry dates prevents monotone decay and forces overshooting, producing oscillation at a frequency set by the market, not the shock.

The empirical evidence supports this isolation. Geographic concentration—a market-structure variable—predicts cycle frequency across 16 manufacturing industries at the 1% level. After large historical shocks, manufacturing and oil display deterministic oscillations at different structural frequencies (5.0 and 3.3 years), ruling out shock-driven explanations that would produce a common cycle length. The two tests are complementary: the cross-section identifies the market-structure channel, the post-shock periodicity identifies the oscillator mechanism.

More broadly, the timing search framework suggests that the spectral properties of industry-level fluctuations are not inherited from aggregate shocks or policy rules but are structural objects determined by how firms in an industry interact. This opens the door to a market-structure theory of business cycle heterogeneity, where the question is not just why economies fluctuate but why different parts of the economy fluctuate at different rates.

References

Beaudry, P., D. Galizia, and F. Portier (2020): "Putting the Cycle Back into Business Cycle Analysis," *American Economic Review*, 110, 1–47.

Beaudry, P. and F. Portier (2006): "Stock Prices, News, and Economic Fluctuations," *American Economic Review*, 96, 1293–1307.

Brock, W. A., W. D. Dechert, J. A. Scheinkman, and B. LeBaron (1996): "A Test for Independence Based on the Correlation Dimension," *Econometric Reviews*, 15, 197–235.

Burdett, K. and D. T. Mortensen (1998): "Wage Differentials, Employer Size, and Unemployment," *International Economic Review*, 39, 257–273.

Christiano, L. J., M. Eichenbaum, and C. L. Evans (2005): "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy," *Journal of Political Economy*, 113, 1–45.

Combes, P.-P., G. Duranton, L. Gobillon, D. Puga, and S. Roux (2012): "The Productivity Advantages of Large Cities: Distinguishing Agglomeration from Firm Selection," *Econometrica*, 80, 2543–2594.

Cooper, R. W. and J. C. Haltiwanger (2006): "On the Nature of Capital Adjustment Costs," *Review of Economic Studies*, 73, 611–633.

Dunne, T., M. J. Roberts, and L. Samuelson (1988): "Patterns of Firm Entry and Exit in U.S. Manufacturing Industries," *RAND Journal of Economics*, 19, 495–515.

Duranton, G. and D. Puga (2004): "Micro-Foundations of Urban Agglomeration Economies," in *Handbook of Regional and Urban Economics*, ed. by J. V. Henderson and J.-F. Thisse, Elsevier, vol. 4.

Ellison, G. and E. L. Glaeser (1997a): "Geographic Concentration in U.S. Manufacturing Industries: A Dartboard Approach," *Journal of Political Economy*, 105, 889–927.

——— (1997b): "Geographic Concentration in U.S. Manufacturing Industries: A Dartboard Approach," *Journal of Political Economy*, 105, 889–927.

Ellison, G., E. L. Glaeser, and W. R. Kerr (2010): "What Causes Industry Agglomeration? Evidence from Coagglomeration Patterns," *American Economic Review*, 100, 1195–1213.

Fisher, R. A. (1929): "Tests of Significance in Harmonic Analysis," *Proceedings of the Royal Society of London. Series A*, 125, 54–59.

Grandmont, J.-M. (1985): "On Endogenous Competitive Business Cycles," *Econometrica*, 53, 995–1045.

Hopenhayn, H. A. (1992): "Entry, Exit, and Firm Dynamics in Long Run Equilibrium," *Econometrica*, 60, 1127–1150.

Jovanovic, B. (1982): "Selection and the Evolution of Industry," *Econometrica*, 50, 649–670.

Krugman, P. (1991): "Increasing Returns and Economic Geography," *Journal of Political Economy*, 99, 483–499.

Kydland, F. E. and E. C. Prescott (1982a): "Time to Build and Aggregate Fluctuations," *Econometrica*, 50, 1345–1370.

——— (1982b): "Time to Build and Aggregate Fluctuations," *Econometrica*, 50, 1345–1370.

Lucas, Robert E., J. (1987): *Models of Business Cycles*, Oxford: Basil Blackwell.

McCall, J. J. (1970): "Economics of Information and Job Search," *Quarterly Journal of Economics*, 84, 113–126.

Ohanian, L. E. (2001): "Why Did Productivity Fall So Much During the Great Depression?" *American Economic Review*, 91, 34–38.

Petrongolo, B. and C. A. Pissarides (2001): "Looking into the Black Box: A Survey of the Matching Function," *Journal of Economic Literature*, 39, 390–431.

Pissarides, C. A. (2000): *Equilibrium Unemployment Theory*, Cambridge, MA: MIT Press, 2nd ed.

Rosenthal, S. S. and W. C. Strange (2004): "Evidence on the Nature and Sources of Agglomeration Economies," in *Handbook of Regional and Urban Economics*, ed. by J. V. Henderson and J.-F. Thisse, Elsevier, vol. 4.

Stigler, G. J. (1961): "The Economics of Information," *Journal of Political Economy*, 69, 213–225.

Syverson, C. (2004): "Market Structure and Productivity: A Concrete Example," *Journal of Political Economy*, 112, 1181–1222.

White, H. (1980): "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity," *Econometrica*, 48, 817–838.